Automated Gas Detection and Ventilation Control System

An ESP32-based safety system that continuously monitors combustible gas (MQ-2) and air quality / VOC levels (MQ-135), automatically activates an exhaust fan and audible alarm when thresholds are exceeded, and serves a live SCADA dashboard over Wi-Fi.

Hardware Components

Component	Role
ESP32 DevKit V1	Main controller
MQ-2 sensor	Detects LPG, propane, methane, smoke
MQ-135 sensor	Detects VOCs, CO_2, ammonia, general air quality
Relay module	Switches the exhaust fan on/off
Buzzer (NPN-driven)	Audible alarm in DANGER state
Red LED	Visual alarm indicator
Green LED	Safe / heartbeat indicator
Alarm-reset button	Silences the buzzer without stopping the fan
16×2 I2C LCD	Local status display

Pin Map

GPIO	Function
34	MQ-2 analogue input (via 10 kΩ/20 kΩ divider)
35	MQ-135 analogue input (via 10 kΩ/20 kΩ divider)
26	Relay (fan)
25	Buzzer (NPN base via 1 kΩ)
27	Red LED (220 Ω series)
14	Green LED (220 Ω series)
13	Alarm-reset button (INPUT_PULLUP, to GND)
21 / 22	I2C SDA / SCL for LCD

The controller **EN** pin is wired to a hardware reset button — no firmware needed for that.

System States

The system moves through four states. **The highest severity across both sensors always wins.**

State	Condition	Fan	Buzzer	Red LED	Green LED
WARMING UP	First 3 minutes after power-on	Off	Off	Off	Slow heartbeat blink
SAFE	Both sensors below WARNING threshold	Off	Off	Off	Steady on
WARNING	Either sensor WARNING threshold	On	Off	Slow blink	Off
DANGER	Either sensor DANGER threshold	On	Pulsing	Steady on	Off

A **hysteresis band of 120 ADC counts** prevents the system from chattering at a threshold boundary — the reading must drop 120 counts *below* a threshold before the state steps back down.

Default Alarm Thresholds (12-bit ADC, 0 – 4095)

Sensor	WARNING	DANGER
MQ-2	1200	2000
MQ-135	1300	2100

> These are starting points. See **Calibration** below for how to tune them to your environment.

LCD Display

The LCD shows different information depending on the current state.

During Warm-up (first 3 minutes)

```
WARMING UP  42s
Alarms inhibitd
```

Field	Meaning
`42s`	Seconds remaining until sensors are ready and alarms become active
`Alarms inhibitd`	Fixed message — no alarms will trigger during this period

Normal Operation

```
WARNING F:ON M
G2: 847 G135:1204
```

Line 1 — Status bar

Field	Values	Meaning
State label	`SAFE` / `WARNING` / `DANGER!`	Current system state
`F:ON` / `F:OFF`	ON or OFF	Whether the exhaust fan is currently running
`M` (trailing)	Present or absent	Alarm has been **muted** by the operator pressing the reset button. The fan keeps running; only the buzzer is silenced. Clears automatically when the air returns to SAFE.

Line 2 — Raw sensor readings

Field	Meaning
`G2:` followed by a 4-digit number	Raw 12-bit ADC reading from the **MQ-2** sensor (LPG / propane / smoke). Range 0 – 4095. Compare against WARNING = 1200, DANGER = 2000.

Field	Meaning
`G135:` followed by a 4-digit number	Raw 12-bit ADC reading from the **MQ-135** sensor (VOC / air quality). Range 0 – 4095. Compare against WARNING = 1300, DANGER = 2100.

Higher numbers mean higher gas concentration. Clean-air baseline is typically 350 – 600 with the recommended voltage divider.

SCADA Web Dashboard

Once connected to Wi-Fi, the ESP32 serves a dashboard at the IP address shown on the LCD during boot. Open that address in any browser on the same network.

The dashboard auto-refreshes every 700 ms. All fields are **read-only** — the dashboard monitors only.

State Banner (top coloured panel)

Colour	State	Message
Blue	SENSOR WARM-UP	Heaters stabilising — alarms inhibited
Green	SAFE	Atmosphere clear — ventilation idle
Amber	WARNING	Gas detected — ventilation running
Red (pulsing)	DANGER	EVACUATE AREA — full ventilation and alarm active

Sensor Cards

Each sensor card shows:

Field	Meaning
% of danger level (large number)	The raw reading expressed as a percentage of the DANGER threshold. 100 % means the DANGER threshold has been reached. Values above 100 % are capped visually at 100 % on the bar but the number can go up to 150 %.
Progress bar	Colour-coded fill: green below 60 %, amber 60 – 99 %, red at 100 %+
`RAW`	The actual 12-bit ADC value (same as `G2:` / `G135:` on the LCD)
`W … / D …`	The configured WARNING and DANGER thresholds for that sensor

Status Tiles

Tile	Meaning
Exhaust Fan	RUNNING (green dot) or STOPPED
Audible Alarm	OFF / SOUNDING (red dot) / SILENCED (amber dot — buzzer muted by operator but alarm condition still active)
Controller Uptime	Time since the ESP32 last powered on, formatted as `hh:mm:ss` or `Xd hh:mm:ss`
Signal / Heap	Wi-Fi signal strength in dBm, and free RAM on the ESP32 in kilobytes

Event Log

A rolling list of the last 12 system events (state changes, fan start/stop, alarm silenced, Wi-Fi status, etc.) with a timestamp showing how long ago each event occurred.

Browser Audio Alerts

If you tap the yellow bar at the bottom of the page to grant permission, the browser will: - Play a **siren** when the state enters DANGER. - Play a **chirp** when the state enters WARNING. - Show a **desktop notification** for WARNING and DANGER transitions.

Calibration

1. Flash the firmware and open **Serial Monitor** at 115 200 baud.
2. Allow the sensors to warm up fully (3 minutes — watch the LCD countdown or Serial output).
3. Note the stable clean-air readings for both sensors — this is your **baseline**.
4. Set `MQ2_WARNING` and `MQ135_WARNING` to roughly **2× − 3× the baseline**.
5. Hold an unlit gas lighter near each sensor, note the peak reading, and set `MQ2_DANGER` / `MQ135_DANGER` comfortably below that peak.
6. Adjust `HYSTERESIS` if the outputs chatter at the boundary.

Wi-Fi Setup

The firmware connects to an existing hotspot (e.g. a phone hotspot) as a **station** (client). Set your hotspot credentials in the sketch:

```
const char* WIFI_SSID     = "your-hotspot-name";
const char* WIFI_PASSWORD = "your-hotspot-password";
```

If Wi-Fi fails to connect within 25 seconds the system continues in **standalone mode** — all local alarms (fan, buzzer, LEDs, LCD) remain fully active; only the web dashboard is unavailable.

Libraries Required

Install via Arduino Library Manager:

- **LiquidCrystal I2C** by Frank de Brabander

The following ship with the ESP32 board package and need no separate install: `WiFi.h`, `WebServer.h`, `Wire.h`

Board settings: ESP32 Dev Module · Upload speed 921600 · Serial monitor 115200 baud